

Quiz 3 Practice Problems

10-606

October 3, 2026

1 Matrix Calculus and Optimization

1. Let $f(x) = (Ax + b)^\top (Ax + b)$ where $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$.
 - (a) Compute $\nabla_x f(x)$.
 - (b) Compute the Hessian $H(f) = \nabla_x^2 f(x)$.
2. Let $X \in \mathbb{R}^{n \times m}$ and $Y \in \mathbb{R}^{m \times n}$.
 - (a) The trace of a square matrix A , denote $\text{Tr}(A)$, is the sum of its diagonal elements. Show that $\text{Tr}(XY) = \sum_{i=1}^n \sum_{j=1}^m x_{ij} y_{ji}$.
 - (b) Show that $\nabla_X \text{Tr}(XY) = Y^\top$.
3. For $\theta, y \in \mathbb{R}^K$ such that $\sum_{i=1}^K y_i = 1$, let

$$f(\theta) = - \sum_{i=1}^K y_i \log \left(\frac{e^{\theta_i}}{\sum_{j=1}^K e^{\theta_j}} \right).$$

What is $\nabla_\theta f(\theta)$?

4. Let $h(w) = \log(1 + \exp(-y w^\top x))$ with $x \in \mathbb{R}^d$ and $y \in \{-1, 1\}$. Compute $\nabla_w h(w)$.
5. For a square matrix $A \in \mathbb{R}^{n \times n}$ and vector $x \in \mathbb{R}^n$, let $f(x) = \frac{x^\top A x}{x^\top x}$. Compute $\nabla_x f(x)$.
6. Let $f_1, \dots, f_n$ be convex functions from $\mathbb{R}^n$ to $\mathbb{R}$.
 - (a) Show that αf_1 is convex for any scalar $\alpha \geq 0$.
 - (b) Show that $\sum_i \alpha_i f_i$ is convex for nonnegative real numbers α_i .
 - (c) Is $\sum_i \alpha_i f_i$ for arbitrary real numbers α_i ?

7. Let $f(x) = \frac{1}{2}x^\top Qx - c^\top x$ with

$$Q = \begin{bmatrix} k & 1 \\ 1 & k \end{bmatrix}, \quad c = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

for some $k \in \mathbb{R}$.

- (a) Compute $\nabla f(x)$.
- (b) From $x_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, take one gradient descent step with step size $\eta = 0.2$. What is x_1 ?

2 Probability

1. You roll two fair dice.
 - (a) $P(\text{sum} = 7)$
 - (b) $P(\text{at least one six})$
 - (c) Are the events in (a) and (b) independent?
2. From a standard 52-card deck, draw two cards without replacement.
 - (a) What is the probability both are hearts?
 - (b) What is $P(\text{second is a heart} \mid \text{first is red})$?
3. Let $X \sim \text{Bernoulli}(p)$ (i.e., X is 1 with probability p , 0 with probability $1 - p$) and define $Y = 1 - X$.
 - (a) Write the joint pmf $P(X, Y)$.
 - (b) Are X and Y independent?
4. Flip a fair coin three times. Let X be the number of heads, and $Y = \mathbf{1}\{\text{first flip is H}\}$.
 - (a) Write the joint pmf $P(X, Y)$.
 - (b) Compute $P(X = 3 \mid Y = 1)$.
5. Show that

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)},$$

if $P(B) \neq 0$. This is known as Bayes' theorem.

6. An urn contains 5 red, 3 blue, and 2 green balls. Two balls are drawn *without replacement*. What is:
 - (a) The probability both are red.
 - (b) The probability the two balls are the same color.
 - (c) The probability the second ball is blue, given the first was green.
7. This is a classic phenomenon known as the “birthday paradox.” Suppose there are 23 people in a room. Let’s assume their birthdays are uniformly distributed across the days of the year. We’re going to show that the probability that two people share a birthday is more than 50%.
 - (a) Let A be the event that some two people in the room share a birthday. Express this probability in terms of the chances that no two people share a birthday.
 - (b) Label the 23 people from 1 to 23. Let E_k be the event that person k does not share a birthday with person 1 through $k - 1$. What is $P(E_k | E_1 \cap E_2 \cap \cdots \cap E_{k-1})$?
 - (c) Express $P(\cap_{i=1}^{23} E_i)$ as a product of terms that look like $P(E_k | \cap_{j < k} E_j)$. (Recall that $\cap_{j < k} E_j$ is just a concise way of writing $E_1 \cap E_2 \cap \cdots \cap E_{k-1}$.)
 - (d) What is $P(A)$?